

git configuration(git settings)

* **Purpose of git Configuration:** To customize Git behavior.

* **Levels of Configuration(Configuration Scope)**

-> In git we have three Levels for configuration

1. Local Level Configuration
2. System Level Configuration
3. Global Level Configuration

Q. What is mean by System Level Configuration?

- > System Level Configuration means Configuration applicable for every user on the system.
- > On Linux operating, system level configuration is stored in `'/etc/gitconfig'` file.
- > In windows operating, system level configuration is stored in `'C:\Program Files\Git\etc\gitconfig'` file

Q. What is mean by Local Level configuration?

- > Local Level Configuration means configuration applicable only for specific repository
- > It's stored in the `.git/config` file in that repository.

Q. What is mean by Global Level Configuration?

- > Global Level Configuration means Configuration applicable for a specific user.
- > It's stored in the user's home directory
- > In windows it stored at below location
c drive-> Users -> User Name -> gitconfig file

git config command

- > **git config** command is used to view the configuration and to perform the configuration settings

View the Configuration Settings

-> To view the configuration settings, use "git config" command with --list option and specify the level of configuration

1. View System level Configuration

-> git config --list --system

2. View Local(repository specific) Configuration

-> git config --list --local

3. View Global(User Specific) Configuration

-> git config --list --global

Q. git config --list --system shows system level configuration,
git config --list --local shows repository level configuration
git config --list --global show user specific configuration
But if we don't specify the scope and execute "git config --list"
command then also we are getting configuration details,
here,question is "git config --list" command shows which level
configuration?

-> It is showing sytem + global level configuration

Note: To exit from "git config" command press "q" (q means quite)

Setting Configuration

1. Set Username and Email

-> These are the two most important configurations, because Git uses username and email in commit history

Global Level

1. git config --global **user.name** "github account user name"
2. git config --global **user.email** "email address here"

Local Level

1. git config --local **user.name** "github account user name"
2. git config --local **user.email** "email address here"

2. Set Default Branch Name

-> When we create a new repository then its initial branch name is "**master**"
But You can configure Git to use a different default branch name (e.g., main).

```
git config --global init.defaultBranch main
```

3. Configure Line Endings

-> Some time we get warning "in working copy of ".gitignore", LF will be replaced by CRLF, next time git touches"

What is meaning of this warning?

We have three types of Line break

- 1) **CR** -> Carriage Return. denoted by "\r"
- 2) **LF** -> Line Feed. denoted by "\n"
- 3) **CRLF** -> Carriage Return Line Feed. denoted by '\r\n'

-> Windows uses **CRLF("\r\n")** to denote new line character while Linux uses **LF("\n")** to denote new line character.

-> When developes using different operating system then line ending must be handled correctly. So handle line ending you need to perform line ending configuration

-> For compatibility across operating systems (especially between Windows and Unix-based systems), you can configure how Git handles line endings:

1. Automatically convert line endings to LF on commit and back to CRLF on checkout (for Windows users)

`git config --global core.autocrlf true`

2. Use LF on commit, and don't change anything on checkout (for Linux/macOS users)

`git config --global core.autocrlf input`

3. Disable line ending conversion

`git config --global core.autocrlf false`